

Preference, Choice and Demand

Advanced Microeconomics 1: Economic Decision Making. Lecture 1

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What are we doing here?

Objective

Academic economists do two things:

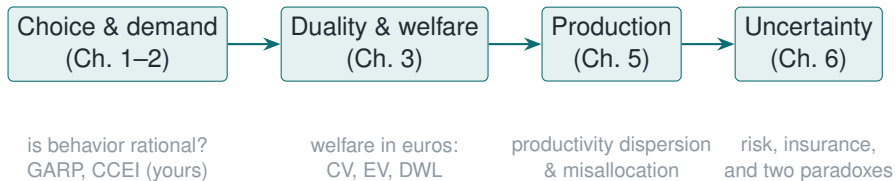
(a) try to understand how the world works

(b) design policy that makes people better off

The formal tools we develop in this course tell us...

- what we can learn from the data under exactly what assumptions
- how to test the assumptions
- what breaks if the assumptions fail

This course



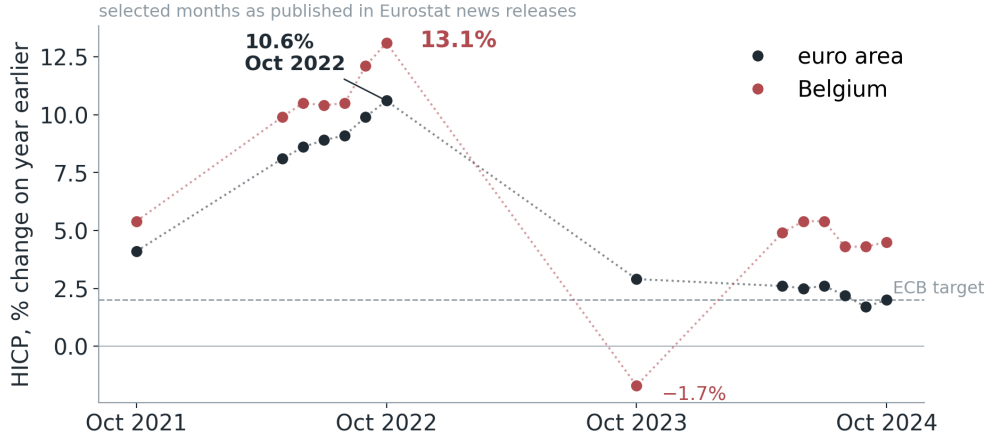
Begin with a quiz

https://lmilsom.github.io/advmico1_decisions/micro1_site/quiz.html

- Remember your ID!
- We will be analysing this data in the practical class.

Motivation: October 2022

Something happened while you were in secondary school



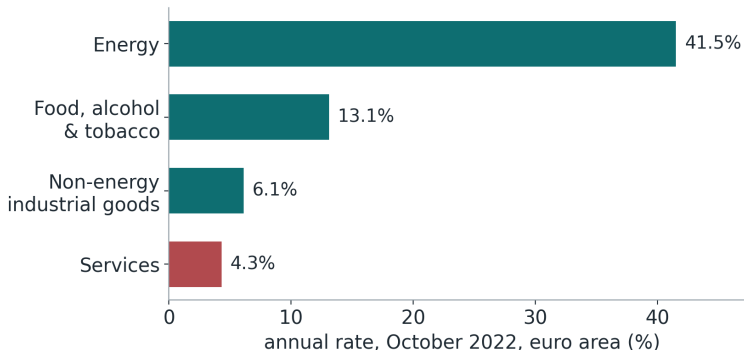
Source: Eurostat (prc_hicp_manr), news releases 130/2022 (17 Nov 2022) and 19 Nov 2024.

One country, one number

October 2022, Belgium: inflation 13.1% — the highest in thirty years.

- One number, for eleven million people.
- By law, that one number matters enormously here: Belgian wages, pensions and benefits are **automatically indexed** to a price index.
- So the number is not a description. **It is a transfer.** If it is wrong for you, you are over- or under-compensated — every month.

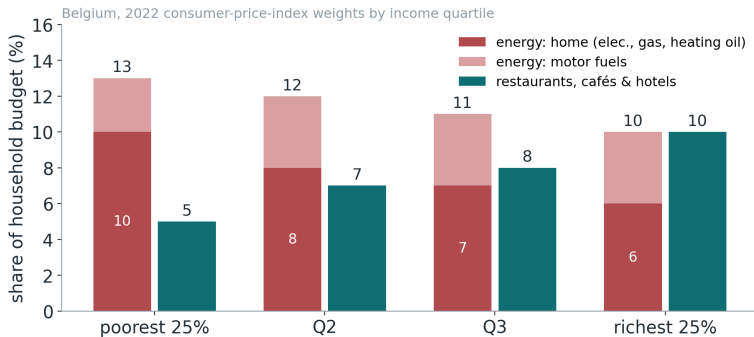
But prices did not rise together



Source: Eurostat (prc_hicp_manr), news release 130/2022, 17 Nov 2022.

Energy nearly $\times 10$ faster than services. In Belgium over 2022: gas +109%, electricity +57% (NBB). “Inflation” was not one price change — it was a violently *relative* price change.

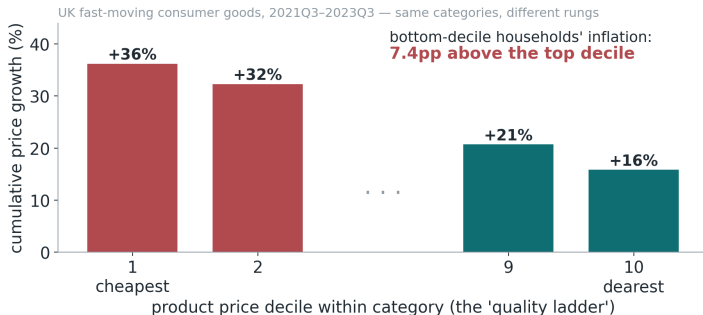
And budgets are not the same



Source: NBB Annual Report 2022, Table 3.2 (Statbel, Household Budget Survey; NBB calculations).

Channel one, visible in any budget survey: the categories that exploded weigh most where budgets are tightest — *“same storm, different boats”* (Capéau et al., Leuvense Economische Standpunten 2022, written down the corridor). There is a second channel, and official statistics cannot see it.

Within categories, it was worse: cheapflation



Source: Chen, Levell & O'Connell (2025), IFS Working Paper 25/21, Fig. 2.1 and §1 (Kantar Take Home Purchase Panel).

Cheap variants rose nearly twice as fast as premium ones, *inside the same categories*. Poorer households live on the low rungs, so their inflation ran away even holding categories fixed: over half of UK inflation dispersion was *within* categories (Chen, Levell & O'Connell 2025). Any category-level index — including today's π^h — is a **floor** on the inequality.

So: whose inflation was 13.1%?

- If price changes differ across goods, and budgets differ across households, then **households faced different inflation rates**.
- This is measured. Jaravel (QJE 2019; Annual Review 2021): in US scanner data, lower-income households' inflation runs persistently higher — on the order of half a percentage point *per year*, compounding. Argente & Lee (JEEA 2021): the gap widens sharply in crises.
- 2022 was the crisis pattern at maximum volume: an energy shock is close to a targeted tax on small budgets. The NBB's own accounting: lower-income households outside the social tariff faced *above-average* inflation; the highest incomes, the lowest.

The questions of this block

(a) understand: who did the 2022 price shock actually hurt — and by how much?

(b) policy: is automatic indexation — one basket, one number — the right compensation? Who wins and who loses from it?

- These sound like statistics questions. They are not: “hurt” means *worse off by the household’s own standard* — and that phrase will force every piece of theory in the next four lectures into existence, one piece at a time.
- Everything we define exists because these two questions demand it. If a definition ever appears unmotivated, stop me.

What answering this requires

Nothing on this list is introduced for its own sake. Each line exists because the question demands it.

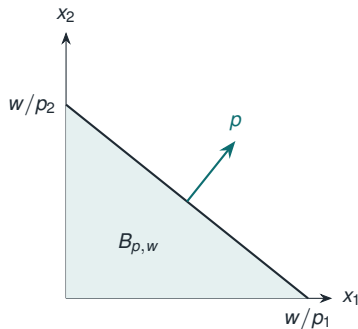
1. **Formalise** what we observe: prices, budgets, demand. today
2. **Describe** without assumptions: whose prices rose, whose budgets are exposed. today
3. **Define the ideal answer**: “as well off as before,” in euros, by *your own* standard. L2
4. **Connect it to data**: when does observed demand determine that answer? L3–L4
5. **Test** the load-bearing assumption: that choices reveal a stable “better off.” today → practical
6. **Audit**: if the assumptions are wrong, which way is the compensation wrong? L4

Formalise: demands, prices, wealth

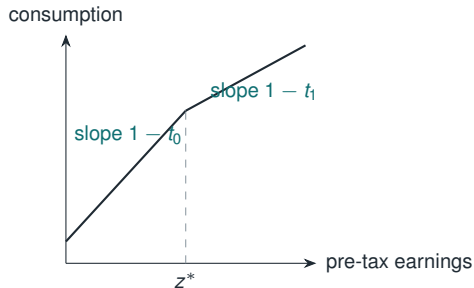
- L commodities, described by a consumption set $X = \mathbb{R}_+^L$.
- L prices $p \in \mathbb{R}_{++}^L$.
- **Walrasian budget set**, $p \gg 0$, $w > 0$:

$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$

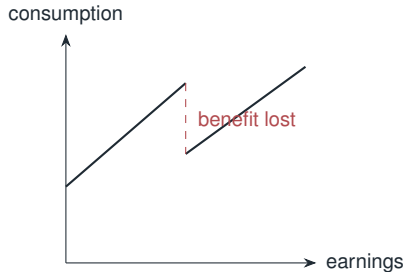
2022 in this notation: p moved — a lot, and very unevenly across its L entries.



Kink (e.g. progressive tax)



Notch (benefit cliff)



Social tariffs for energy — extended from roughly 500 000 to about one million Belgian households in 2021 (NBB) — are exactly a notch: cross an income line, lose the tariff.

Definition · Walrasian (Marshallian) demand

$x(p, w) \subseteq B_{p,w}$: the chosen bundle(s) at each price–wealth pair; assume single-valued and continuous.

Two assumptions:

1. **Homogeneity of degree zero:** $x(\alpha p, \alpha w) = x(p, w)$ — only relative prices and real wealth matter; “no money illusion.” Practical use: normalize.
File this claim away. It is an assumption about how people behave *during inflations*. We return to it, armed, in Lecture 4.
2. **Walras’ law:** $p \cdot x(p, w) = w$ — full expenditure.

The four formal objects

- Consumption set: $X \in \mathbb{R}_+^L$.
- Prices: $p \in \mathbb{R}_{++}^L$.
- Budget set: $B_{p,w} = \{x \in X : p \cdot x \leq w\}$.
- Demand: $x(p, w) \subseteq B_{p,w}$.

Enough to state the events of 2022 precisely. Not yet enough to say who was hurt and by how much.

What can we already learn: commodities

Describing commodities

Comparative statics: how does behavior change when the economic parameters change.

How x_l varies with p_k , w tells us about the properties of commodity l .

No rationality anywhere in this section — just derivatives of observable demand. It will take us surprisingly far into the inflation question.

Changing wealth — some definitions

Keep p constant at $\bar{p}$, vary w , then ...

- $x(\bar{p}, w)$ is the Engel curve
- $E_{\bar{p}} = \{x(\bar{p}, w) : w > 0\}$ is the wealth expansion path
- $\frac{\partial x_l(p, w)}{\partial w}$ is the wealth effect for commodity l
- $\epsilon_{lw}(p, w) = \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)}$ is the elasticity of demand w.r.t. wealth.
- $b_l(p, w) = \frac{p_l x_l(p, w)}{w}$ is the budget share of l .

Changing wealth — categorising commodities

A commodity l is a ...

- normal commodity if $\frac{\partial x_l(p, w)}{\partial w} \geq 0$
- inferior commodity if $\frac{\partial x_l(p, w)}{\partial w} < 0$
- luxury commodity if $\epsilon_{lw}(p, w) \geq 1$, budget share increases in wealth
- necessity commodity if $\epsilon_{lw}(p, w) < 1$, budget share decreases in wealth

Engel (1857): food's share falls with wealth — robust across countries and centuries ($\sim 50\%$ in poor countries, $\sim 15\%$ in rich ones). Within Belgian quartiles, home energy does the falling ($10\% \rightarrow 6\%$) and restaurants the rising ($5\% \rightarrow 10\%$): the motivation chart is Engel's law running *both* directions — and 2022 punished the necessity end.

Changing prices — some definitions

- $\frac{\delta x_l(p, w)}{\delta p_k}$ is the price effect of p_k on commodity l .
- $\epsilon_{lk}(p, w) = \frac{\delta x_l(p, w)}{\delta p_k} \frac{p_k}{x_l(p, w)}$ is the elasticity of demand for l with respect to p_k .
 - ϵ_{ll} = own price elasticity
 - ϵ_{lk} = cross price elasticity

Changing prices — categorising commodities

- Demand for l is elastic if $\epsilon_{ll}(p, w) \leq -1$.
- Demand for l is inelastic if $-1 < \epsilon_{ll}(p, w) \leq 0$.
- Demand for l is a Giffen good if $\epsilon_{ll}(p, w) > 0$.

Home energy in the short run: about as inelastic as demand gets — you cannot quickly stop heating. Inelastic + large budget share is exactly the profile that makes a price shock hurt.

First payoff: your inflation rate

Let household h have budget shares b_l^h . For small price changes, the inflation *that household faces* is

$$\pi^h = \sum_{l=1}^L b_l^h \pi_l \quad (\text{share-weighted price growth}).$$

- Different $b^h \Rightarrow$ different π^h . The components chart $\times$ the quartile-weights chart = the Jaravel fact. **Zero assumptions used.**
- The official index does this with *aggregate expenditure* weights — so each euro, not each household, gets one vote. A rich household's basket counts more. (“Plutocratic” vs. “democratic” index — EC1.)
- Lurking question: is share-weighting even the right formula? It ignores that people *substitute away* from what got expensive. Answering that properly needs everything in the next three lectures.

The load-bearing assumption

Why we now need an assumption

“Worse off” — by whose standard?

- The compensation question is about *welfare*: each household's own ranking of situations. Rankings are not observable. Choices are.
- So the entire program rests on one bridge: **choices reflect a stable ranking**. That is the rationality hypothesis — not a creed, an assumption we now make precise, and then *test*.
- The stakes are concrete: if the bridge holds, Lectures 2–4 convert each household's demand into a euro answer. If it fails, the compensation apparatus is built on sand — so testing it first is due diligence, not philosophy.

It is testable: consistency scores in a large Dutch panel are heterogeneous and correlate with wealth, education, age (Choi, Kariv, Müller & Silverman, AER 2014). The same test scores GPT *more* consistent than most humans (Chen et al., PNAS 2023). In a week: you.

Preferences

Definition · preference relation

$\succsim$ is a binary relation on X ; $x \succsim y$ reads “ x is at least as good as y .”

- x is better than y : $x \succ y \Leftrightarrow x \succsim y$ and not $y \succsim x$.
- x is indifferent to y : $x \sim y \Leftrightarrow x \succsim y$ and $y \succsim x$.

This is the “standard” the question needs: household h ’s own ranking, $\succsim_h$.

Rationality

Definition · rationality

$\succsim$ is *rational* if $\forall x, y, z \in X$ it is:

complete: $x \succsim y$ or $y \succsim x$

transitive: $x \succsim y \succsim z \Rightarrow x \succsim z$

Definition · utility representation

$u : X \rightarrow \mathbb{R}$ represents $\succsim$ if $x \succsim y \iff u(x) \geq u(y)$.

Why we want a u : euros are numbers. “How much worse off” must eventually be arithmetic, and arithmetic needs a function — built properly in Lecture 2.

Proposition · MWG 1.B.2

If a utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$, then $\succsim$ is rational.

► proof

$\succsim$ inherits the properties of $\geq$ on $\mathbb{R}$ through the representation.

- The converse is the expensive direction: rational $\Rightarrow$ representable needs more (continuity — Lecture 2). For now, one direction is free.

Testing the assumption on the data we have

What would rationality look like in the data?

Our data: pairs (p^t, x^t) — prices faced, bundle chosen — or an estimated demand function $x(p, w)$. *No rankings anywhere.*

- The bridge, used in reverse: choosing x when y was **affordable** reveals $x \succsim y$. Each observation is a small confession about the ranking.
- Rationality then has an observable footprint: the confessions must not contradict each other.

Definition · WARP for demand functions (MWG 2.F.1)

$$p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w) \implies p' \cdot x(p, w) > w'.$$

Read: if the bundle chosen at (p', w') was affordable at (p, w) but passed over, then when it *was* chosen, the other bundle must have been out of reach. Pairs of observations must not accuse each other.

Is pairwise consistency the whole test? No.

goal (a): understand

Three observed budget-set choices ($L = 3$ goods):

t	p^t	x^t	$w^t = p^t \cdot x^t$
1	(1, 1, 2)	(2, 0, 0)	2
2	(2, 1, 1)	(0, 2, 0)	2
3	(1, 2, 1)	(0, 1, 1)	3

- Obs. 1 reveals $x^1 \succsim x^2$. Obs. 2 reveals $x^2 \succsim x^3$. Obs. 3 reveals $x^3 \succ x^1$ — affordable *with money left over*. A cycle: no coherent ranking exists, so “how much worse off” has no answer for this household.
- Yet every *pair* is clean: each reversal is unaffordable. **WARP is satisfied**. A pairwise axiom cannot see a three-link disease.
- (With $L = 2$ this cannot happen — for two goods WARP already rules out all cycles. Relevant in a week: your quiz has two goods.)

Definition · GARP

Write $x^t R x^s$ iff $p^t \cdot x^s \leq w^t$ (directly revealed preferred), P for strict. **GARP**: no chain $x^{t_1} R x^{t_2} R \dots R x^{t_n}$ with $x^{t_n} P x^{t_1}$.

- WARP is the $n = 2$ case; GARP follows the confessions around loops of any length.
- **Afriat's Theorem** (1967; proved in Lecture 4): a finite dataset satisfies GARP *iff* it is rationalizable by a continuous, increasing, concave utility. For finite data, **utility maximization = GARP**. Nothing more, nothing less.

Why this matters for the inflation question: in Lecture 4 we compensate households by turning their *demand* back into their *preferences*. GARP is the certificate that this inversion is allowed — household by household. No certificate, no euro answer. Thursday week you compute the certificate for everyone in this room.

The compensation problem, first pass

Compensated price changes

Back to goal (b). A price change does two things: changes relative prices, and changes real wealth.

Definition · (Slutsky) compensated price change

From (p, w) with $x = x(p, w)$, move to p' and set $w' = p' \cdot x$: wealth adjusted so *the old bundle remains exactly affordable*. The adjustment $\Delta w = \Delta p \cdot x(p, w)$ is the Slutsky wealth compensation.

- This is not a proof device. It is **indexation policy in miniature**: pay each household enough that its old basket stays affordable at the new prices. Belgium's mechanism is a country-sized Slutsky compensation.
- So the first-pass answer to goal (b) is on the table: compensate by $\Delta p \cdot x$. The next two slides let the theory cross-examine it.

What a compensated household does

Theorem · MWG 2.F.1

Suppose $x(p, w)$ is h.d.0 and satisfies Walras' law. Then x satisfies WARP $\iff$ for every compensated price change,

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad \text{strict if } x(p', w') \neq x(p, w).$$

► proof

- A *law of demand*: a compensated household tilts away from what got relatively dearer ($\Delta p_\ell > 0 \Rightarrow \Delta x_\ell \leq 0$). Uncompensated demand obeys no such law (Giffen); compensation buys the sign.
- In inflation language: hand a household its Slutsky compensation and it will **not** rebuy the old basket — it substitutes. That single fact has a policy consequence, next.

First policy result of the course

Basket indexation over-compensates

Compensate a household so its old bundle x stays affordable ($w' = p' \cdot x$). If it chooses $x' \neq x$, then it chose x' *while x was affordable*: by revealed preference, the household is **at least as well off as before — with the old standard of living still in reach**. Full basket indexation is (weakly) more than needed; the true cost of staying as-well-off is lower, because people substitute.

- This is the **substitution bias** of fixed-basket indices — the oldest criticism of the CPI, derived here in three lines.
- Note what the argument leaned on: *revealed preference* — the load-bearing assumption, used the very first time we made a welfare claim. No rationality, no result.
- How *much* over-compensation? That is a magnitude, and it needs the machinery of Lectures 2–3. The next object is its measuring device.

The over-payment above is exactly what the fixed basket ignores: *substitution*. To quantify it we need the derivative of demand **along a compensated change** — from observables only.

Take dp small, compensate $dw = x \cdot dp$; then $dx = [D_p x + D_w x x^T] dp$ and the law of demand reads $dp \cdot dx \leq 0$.

Definition · Slutsky substitution matrix

$$S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^T, \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k.$$

- Every entry is estimable from demand data alone — **no utility function anywhere**. S is where the index's error lives; Lecture 3 turns it into the correction term.

Proposition · MWG 2.F.2–2.F.3

If x is h.d.0, satisfies Walras' law and WARP, then $S(p, w)$ is **negative semidefinite** ($v \cdot Sv \leq 0$) and **singular**: $S p = 0$, $p^T S = 0$.

▶ proof: NSD

▶ proof: singularity

- $s_{\ell\ell} \leq 0$: compensated own-price effects are nonpositive. Corollary: a Giffen good must be inferior.

What WARP does not buy

For $L \geq 3$, WARP does **not** imply S **symmetric** (MWG Ex. 2.F.1). Symmetry is the extra content of *transitivity* — one coherent ranking behind all choices — and it turns out to be the exact condition for *recovering* preferences from demand (Lecture 4). Recovery is what per-household compensation needs: no symmetry, no way back from $x(p, w)$ to $\succsim$, no euro answer.

Where the question stands

Takeaways

- **Stated precisely:** 2022 was a relative-price shock hitting heterogeneous budgets; $\pi^h = \sum_I b_I^h \pi_I$ already follows from the formalism alone.
- **One policy result already:** basket indexation over-compensates — three lines of revealed preference. The size of the over-payment lives in S (Lecture 3).
- **The test is armed:** rationality has an observable footprint — WARP, and for cycles GARP; Afriat makes GARP the *whole* content of the hypothesis on finite data.
- **The gap:** symmetry of S is missing from WARP — precisely the ticket back from demand to preferences (Lecture 2 builds what it buys).

Required: MWG Ch. 2 (all); Example 2.F.1. (Ch. 1 optional.)

Papers: Choi, Kariv, Müller & Silverman (AER 2014), intro + Table 1; Jaravel (QJE 2019), intro; Chen–Levell–O’Connell (IFS 2025). *One more is revealed in Lecture 4.*

Ahead: practical in a week — laptop with working R. Check it starts *this week*.

Appendix: proofs

Proof: utility representation $\Rightarrow$ rationality (MWG 1.B.2)

Recall: u represents $\succsim$ means $x \succsim y \iff u(x) \geq u(y)$.

- *Completeness.* Take any $x, y \in X$. Then $u(x)$ and $u(y)$ are real numbers, and $\geq$ on $\mathbb{R}$ is complete: $u(x) \geq u(y)$ or $u(y) \geq u(x)$. Reading the representation right-to-left, $x \succsim y$ or $y \succsim x$.
- *Transitivity.* Suppose $x \succsim y$ and $y \succsim z$. By the representation, $u(x) \geq u(y)$ and $u(y) \geq u(z)$; transitivity of $\geq$ on $\mathbb{R}$ gives $u(x) \geq u(z)$; right-to-left again, $x \succsim z$. ■

◀ back

Proof: WARP $\Rightarrow$ compensated law (MWG 2.F.1, direction (i))

If $x(p', w') = x(p, w)$ the claim is trivial. So take $x' \equiv x(p', w') \neq x \equiv x(p, w)$ and expand:

$$(p' - p) \cdot (x' - x) = p' \cdot (x' - x) - p \cdot (x' - x).$$

- **First term is zero:** $p' \cdot x' = w'$ (Walras) and $p' \cdot x = w'$ (definition of the compensation). So the old bundle is *exactly affordable* at the new situation.
- **Second term is positive:** x affordable at (p', w') but x' chosen there $\Rightarrow$ (WARP) x' was *not* affordable at (p, w) : $p \cdot x' > w = p \cdot x$. Hence $p \cdot (x' - x) > 0$.
- Together: $(p' - p) \cdot (x' - x) = 0 - (\text{positive}) < 0$. ■

Direction (ii), compensated law $\Rightarrow$ WARP, is tedious and not extremely insightful; see MWG.

Proof: WARP $\Rightarrow$ S negative semidefinite (MWG 2.F.2)

Fix (p, w) , take any direction $v \in \mathbb{R}^L$ and a small price step $dp = \varepsilon v$, *compensated*:
 $dw = x(p, w) \cdot dp$.

- Demand is differentiable, so the induced change in demand is, to first order,

$$dx = D_p x dp + D_w x dw = [D_p x + D_w x x^T] dp = S(p, w) dp.$$

- The change is a compensated price change, so the compensated law of demand (Theorem 2.F.1, which used WARP + h.d.0 + Walras) applies:

$$dp \cdot dx \leq 0 \quad \implies \quad \varepsilon^2 (v \cdot S(p, w) v) \leq 0.$$

- $\varepsilon^2 > 0$ and v was arbitrary: $v \cdot S v \leq 0$ for all v . ■

Proof: $Sp = 0$ and $p^T S = 0$ (from h.d.0 and Walras; MWG 2.E)

Neither part uses WARP — pure accounting.

- **h.d.0:** $x(\alpha p, \alpha w) = x(p, w)$ for all $\alpha > 0$. Differentiate in α at $\alpha = 1$:

$$D_p x p + D_w x w = 0. \quad (\text{E1})$$

- **Walras:** $p \cdot x(p, w) = w$ identically. Differentiate in w , and in p :

$$p^T D_w x = 1, \quad p^T D_p x + x^T = 0. \quad (\text{E2}, \text{E3})$$

- **Right-multiply by p :** $Sp = D_p x p + D_w x (x \cdot p) \stackrel{\text{Walras}}{=} D_p x p + D_w x w \stackrel{(\text{E1})}{=} 0$.

- **Left-multiply by p^T :** $p^T S = p^T D_p x + (p^T D_w x) x^T \stackrel{(\text{E2})}{=} p^T D_p x + x^T \stackrel{(\text{E3})}{=} 0$. ■